
Revisiting ASkewSGD: An Annealed Interval-Constrained Optimization Method for Quantization of Deep Neural Networks

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Abstract

The question of enforcing quantization on the weights and activations in Deep Neural Networks (DNNs) has come to the forefront in recent years due to its relevance to restricted memory and/or computational resources. While low-precision fixed integer values could substantially reduce the memory footprint and latency, inaccurate quantization on model parameters is susceptible to significant accuracy drops. In this paper, we will revisit the Annealed Skewed Stochastic Gradient Descent algorithm (ASkewSGD) proposed by Leconte et al. (2023) for quantization-aware training in deep neural networks [1]. Furthermore, we establish additional convergence guarantees for ASkewSGD by removing the assumption of high-order differentiability of the loss function when the stochastically sampled gradient is replaced with full-batch gradient.¹

1 Introduction

Over the past decade, there has been an increasing awareness that deep learning models have succeeded in achieving vast improvements for tasks such as computer vision, speech recognition and generation, and natural language processing. While the accuracy of these models has significantly improved, the size of these models could be a potential problem when being deployed in resource-limited scenarios. As such, there is a growing interest in the search for efficient design, training, and deployment of neural network models, while allowing optimality trade-offs. These methods include knowledge distillation, micro-architecture design, hardware-architecture co-design, pruning of insensitive neurons, low-rank decomposition, and network quantization [2]. In this paper, we mainly concentrate on network quantization during model training, with weights mapping values from a large set of real numbers to values in a set of discrete values with lower bit widths.

Contributions. The main goal of this work is to improve our understanding of ASkewSGD and its convergence properties. We summarize our contributions as follows:

- We provide new convergence insights for the deterministic version of ASkewSGD, by modifying the remoteness measure in [1], where the deterministic gradient descent eliminates the strong assumption of high-order differentiability of the loss function.
- We evaluate the performance of ASkewSGD on the MNIST and CIFAR-10 dataset and compare it with the state-of-the-art quantization-aware training methods.²

¹Extension of the research work under the supervision of Prof. WAI Hoi To during a summer research internship organized by Faculty of Engineering in 2024. Further extensions for the stochastic case is still in progress. Not to be distributed.

²Related codes are available at <https://github.com/SWongHF/MATH4230>.

2 Formalization

We are interested in solving the optimization problem related to learning a quantized neural network (QNN),

$$\min_{\mathbf{w} \in \mathcal{Q}} \ell(\mathbf{w}), \text{ where } \ell(\mathbf{w}) = \mathbb{E}_{(\mathbf{x}, y) \sim p_{\text{data}}} [L(f(\mathbf{x}, \mathbf{w}), y)], \quad (1)$$

where $f : \mathbb{R}^n \times \mathbb{R}^d \rightarrow \mathbb{R}^m$ models the neural network with the network weights $\mathbf{w} \in \mathbb{R}^d$ and the input $\mathbf{x} \in \mathbb{R}^n$. Here, $L : \mathbb{R}^d \rightarrow \mathbb{R}$ denotes the training loss, $\mathcal{Q} \subset \mathbb{R}^d$ is a discrete, non-convex set of quantization levels, p_{data} is the training distribution.

We do not necessarily aim to solve the above (hard, combinational) optimization problem globally and optimally. Instead, we want to find discrete weights $\mathbf{w} \in \mathcal{Q}$ that remain satisfactory when compared to the non-quantized continuous weights [2]. This circumvents the difficulty as the structure of ℓ could come into aid. If ℓ is reasonably smooth, a closed-by quantized weight of a locally optimal continuous weight will likely yield similar performance.

Notations. We use $\|\cdot\|_2$ to denote the Euclidean norm for vectors. We denote the i -th component of a vector $\mathbf{w}$ as w_i or $[\cdot]_i$. We also use the superscript $\mathbf{w}^{(k)}$ to denote the k -th iterate of the optimization algorithm. We denote a sample (sub)gradient of ℓ at $\mathbf{w}$ as $\widehat{\nabla} \ell(\mathbf{w})$ so that $\mathbb{E}[\widehat{\nabla} \ell(\mathbf{w})] = \nabla \ell(\mathbf{w})$.

3 Previous Works

BinaryConnect. Courbariaux et al. [3] considered an aggressive quantization scheme using binary networks, where $\mathcal{Q} = \{\pm 1\}^d$. The BinaryConnect algorithm updates the weights by the following scheme:

$$\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} - \gamma_k \widehat{\nabla} \ell(\mathbf{P}(\mathbf{w}^{(k)})), \text{ where } [\mathbf{P}(\mathbf{w})]_i = \begin{cases} +1, & w_i \geq 0, \\ -1, & \text{otherwise.} \end{cases} \quad (2)$$

While the above method is deterministic, yet another scheme quantizes the weights stochastically, which is given by:

$$[\mathbf{P}(\mathbf{w})]_i = \begin{cases} +1, & \text{with probability } \sigma(w_i), \\ -1, & \text{with probability } 1 - \sigma(w_i), \end{cases} \quad \text{where } \sigma(x) = \text{clip}\left(\frac{x+1}{2}, 0, 1\right). \quad (3)$$

Despite its empirical success, the convergence of BinaryConnect is not well understood.

ProxQuant. Bai et al. [4] proposed the proximal gradient method for quantization-aware training, which is a variant of the proximal operator

$$\mathbf{w}^{(k+1)} \leftarrow \mathbf{P}\left(\mathbf{w}^{(k)} - \eta_k \widehat{\nabla} \ell(\mathbf{w}^{(k)})\right), \quad (4)$$

where the proximal operator is defined as

$$\mathbf{P}(\mathbf{w}) = \arg \min_{\bar{\mathbf{w}} \in \mathbb{R}^n} \left(\frac{1}{2} \|\mathbf{w} - \bar{\mathbf{w}}\|_2^2 + \lambda R(\bar{\mathbf{w}}) \right), \text{ where } R(\bar{\mathbf{w}}) = \inf_{\hat{\mathbf{w}} \in \mathcal{Q}} \|\hat{\mathbf{w}} - \bar{\mathbf{w}}\|_2.$$

This method is theoretically sound, but the proximal operator is expensive to compute when $\mathcal{Q}$ is large.

4 Preliminaries

Inspired by the success of ProxQuant, Leconte et al. [1] considers a local approximation for the discrete constraints, formulated as a smoothed sequence of interval-constrained optimization problems ($\mathcal{P}_\varepsilon$). In particular, we shed the light on the following optimization problem:

$$\min_{\mathbf{w} \in C_\varepsilon} \ell(\mathbf{w}), \text{ where } C_\varepsilon := \{\mathbf{w} \in \mathbb{R}^n \mid g_{\varepsilon, i}(\mathbf{w}) \geq 0, i = 1, 2, \dots, c\}. \quad (5)$$

To model this problem for the quantization setting, we will need to define the constraints with respect to the quantization set $\mathcal{Q}$. We emphasize that the method supports even the coordinate-wisely defined quantization set $\mathcal{Q}$.

Definition (Quadratic Remoteness Measure of Iterates to the Quantization Set). Let $\varepsilon \in [0, 1]$, $w_i \in \mathcal{Q}_i, i \in [d]$, where $\mathcal{Q}_i = \{q_i^{(1)}, \dots, q_i^{(K_i)}\}$ are sets of quantization values defined coordinate-wise. We define the piecewise function

$$\psi_i(w_i; \varepsilon) := \begin{cases} \varepsilon - (q_i^{(1)} - w_i) & w_i < q_i^{(1)}, \\ \varepsilon - (q_i^{(j-1)} - w_i)(w_i - q_i^{(j)}) & q_i^{(j-1)} \leq w_i < q_i^{(j)}, j = 2, \dots, K_i, \\ \varepsilon - (w_i - q_i^{(K_i)}) & w_i \geq q_i^{(K_i)}, \end{cases}$$

for all $w_i \in \mathbb{R}$.

By setting $g_i(\mathbf{w}) = \psi_i(w_i; \varepsilon)$, we obtain a compact set C_ε , which is also non-empty. Furthermore, as $\varepsilon \rightarrow 0^+$, the set C_ε converges to the quantization set $\mathcal{Q}$.

We describe several classical assumptions on the loss function ℓ and the constraint function g . To employ the mini-batch stochastic gradient, we consider ℓ , the training loss, as $\ell(\mathbf{w}) = 1/N \sum_{j=1}^N \ell_j(\mathbf{w})$.

Assumption 1 (Standing). The Mangasarian-Fromovitz constraint qualification holds, i.e. for all $\mathbf{w} \in \mathbb{R}^d, \exists \mathbf{v} \in \mathbb{R}^d$, such that $\mathbf{v}^\top \nabla g_i(\mathbf{w}) > 0$ for all $i \in I(\mathbf{w}) := \{i \in \mathbb{Z} \mid g_i(\mathbf{w}) \leq 0\}$.

Remark. It is easy to verify the Mangasarian-Fromovitz constraint qualification for our specific case, as the constraint function g_i is coordinate-wisely independent and we can always follow the gradient direction of g_i on the i -th coordinate.

Assumption 2 (Existence of a Minimizer). We assume that ℓ_j is continuously differentiable and has L_{ℓ_j} -Lipschitz continuous gradients. Then, ℓ is $(\sup_{1 \leq i \leq N} L_{\ell_j})$ -Lipschitz continuous.

Remark. Assumption 2 ensures the existence of a minimizer $\mathbf{w}^*$ for the optimization problem by the Weierstrass Theorem, since the loss function ℓ is continuous, and the set C_ε is non-empty, closed and bounded.

Assumption 3 (Decreasing Stepsizes). The stepsizes $(\gamma_k)_{k \geq 0}$ are positive, $\sum_{k=0}^{\infty} \gamma_k = +\infty$, and $\sum_{k=0}^{\infty} \gamma_k^2 < +\infty$.

Remark. Assumption 3 is a standard assumption in stochastic optimization. It ensures that the stepsizes are non-summable and squared summable, which is necessary for convergence.

Assumption 4 (Smooth Loss Function). Furthermore, we assume that ℓ_j is d -times continuously differentiable.

By Assumption 2 alone is enough to well-define the optimization problem. However, we still need some shorthands for the elements in the ASkewSGD algorithm.

Definition (Tangent and Normal Cones). The Clarke's tangent cone of C_ε contains all $\delta \mathbf{x} \in T_{C_\varepsilon}(\mathbf{x})$ if there exists two sequences $\mathbf{x}_j \rightarrow \mathbf{x}, \mathbf{x}_j \in C_\varepsilon, t_j \downarrow 0$ such that $(\mathbf{x}_j - \mathbf{x})/t_j \rightarrow \delta \mathbf{x}$. The normal cone is defined as follows: $N_{C_\varepsilon}(\mathbf{x}) = \{\lambda \in \mathbb{R}^n \mid \lambda^\top \delta \mathbf{x} \leq 0, \forall \delta \mathbf{x} \in T_{C_\varepsilon}(\mathbf{x})\}$.

Lemma 1 (R. Herzog, [5]). Suppose that $\mathbf{x} \in C_\varepsilon$ and the Mangasarian-Fromovitz constraint qualification holds for every $\mathbf{x} \in C_\varepsilon$, then every $\delta \mathbf{x} \in T_{C_\varepsilon}(\mathbf{x})$ satisfies $\nabla g_i(\mathbf{x})^\top \delta \mathbf{x} \geq 0, \forall i \in I(\mathbf{x})$. The converse also holds.

Hence, we can simplify tangent and normal cones of C_ε at $\mathbf{w} \in C_\varepsilon$, given by

$$T_{C_\varepsilon}(\mathbf{w}) = \{\mathbf{v} \in \mathbb{R}^d : \nabla g_i(\mathbf{w})^\top \mathbf{v} \geq 0, \forall i \in I(\mathbf{w})\}, \quad (6)$$

$$N_{C_\varepsilon}(\mathbf{w}) = \left\{ - \sum_{i \in I(\mathbf{w})} \lambda_i \nabla g_i(\mathbf{w}), \lambda_i \in \mathbb{R}_+ \right\}. \quad (7)$$

Furthermore, since the MFCQ holds, the set of stationary points are given by the Karush-Kuhn-Tucker condition:

$$\mathcal{Z}_\varepsilon := \{\mathbf{w} \in C_\varepsilon : \mathbf{0} \in -\nabla \ell(\mathbf{w}) - N_{C_\varepsilon}(\mathbf{w})\}.$$

Now, we approximate and extend the tangent cone “outside” of the feasible set, by $V_{\varepsilon, \alpha} := \{\mathbf{v} \in \mathbb{R}^d : v_i \psi'_i(w_i; \varepsilon) \geq -\alpha \psi_i(w_i; \varepsilon) \text{ for } i \in I(\mathbf{w})\}$.

We are ready to state the ASkewSGD algorithm. The basic idea is to “skew” the search direction in order to force the algorithm to find a minimizer without constraining the sequence $(\mathbf{w}^{(k)})_{k \in \mathbb{N}}$ to the feasible set. We note that this is due to Muehlebach and Jordan’s [6] reformulation of the position constraints into forward “velocity” constraints, which considers a linear and convex approximation of the original feasible set with the objective function satisfying the Mangasarian-Fromovitz condition.

Algorithm 1: ASkewSGD Algorithm for QNN Training

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1 for  $k = 1, 2, \dots$  do
2   Obtain a stochastic gradient  $\widehat{\nabla} \ell(\mathbf{w}^{(k)})$ .
3    $\widehat{\mathbf{v}}^{(k)} \leftarrow \text{clip}(\arg \min_{\mathbf{v} \in V_{\varepsilon, \alpha}(\mathbf{w}^{(k)})} (1/2) \|\mathbf{v} + \widehat{\nabla} \ell(\mathbf{w}^{(k)})\|^2, M_c)$ .
4    $\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} + \gamma_k \widehat{\mathbf{v}}^{(k)}$ .
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In each iteration, we try to find a skewed (stochastic) gradient $\widehat{\mathbf{v}}^{(k)}$ so that

$$\widehat{\mathbf{v}}^{(k)} = \arg \min_{\mathbf{v} \in V_{\varepsilon, \alpha}(\mathbf{w}^{(k)})} \frac{1}{2} \left\| \mathbf{v} + \widehat{\nabla} \ell(\mathbf{w}^{(k)}) \right\|^2, \quad (8)$$

where $\alpha > 0$ is a parameter that controls the trade-off between two objectives: for large α , the emphasis is on the convergence to the feasible set, while for small α , the focus is on reducing the objective function. Furthermore, this optimization problem

$$\arg \min_{\mathbf{v} \in V_{\varepsilon, \alpha}(\mathbf{w})} (1/2) \|\mathbf{v} + \mathbf{u}\|^2$$

has an explicit solution:

$$[\mathbf{s}_{\varepsilon, \alpha}(\mathbf{u}, \mathbf{w})]_i := \begin{cases} -u_i & \text{if } \psi_i(w_i; \varepsilon) > 0 \text{ or } -u_i \cdot \psi'_i(w_i; \varepsilon) \geq -\alpha \psi_i(w_i; \varepsilon) \geq 0, \\ -\alpha \psi_i(w_i; \varepsilon) / \psi'_i(w_i; \varepsilon) & \text{otherwise.} \end{cases}$$

To ensure that $[\mathbf{s}_{\varepsilon, \alpha}(\mathbf{u}, \mathbf{w})]_i$ does not diverge, we apply the clip operator to bound the update direction $\widehat{\mathbf{v}}^{(k)}$ within a fixed range controlled by M_c .

Under mild assumptions, the optimization algorithm has a convergence guarantee summarized by the following theorem, by Leconte et al. [1].

Theorem 1 (Leconte et al., [1]). Assume that Assumptions 2-4 holds, and $0 < \varepsilon \leq \inf_{1 \leq i \leq d} \inf_{1 \leq j < K_i} |q_i^{(j)} - q_i^{(j+1)}|^2 / 4$, where $\{q_i^{(j)}\}$ are the quantization levels, $\ell(\mathbf{w}^{(k)})$ converges and $\lim_{k \rightarrow \infty} d(\mathbf{w}^{(k)}, \mathcal{Z}_\varepsilon) = 0$ almost surely.

Leconte’s proof relies heavily on a “tame” property by Davis et al. [7]. It assumes the Weak Sard’s property of the loss function ℓ ; however, Leconte et al. directly applied the Sard’s Theorem, which utilizes the high-order differentiability. Our work is to remove this strong assumption which is not often realized in the settings of Deep Neural Networks (DNNs).

In the next section, we will also utilize a technical lemma by Leconte et al. [1], which is stated in the following. The proof is deferred to the Appendix.

Lemma 2 (Leconte et al., [1]). Under Assumptions 2-3, it holds that $\limsup_{k \rightarrow \infty} d(\mathbf{w}_k, C_\varepsilon) = 0$ almost surely.

5 New Insights

In this section, we will establish convergence for the deterministic version of ASkewSGD, i.e.

$$\text{Var}(\widehat{\nabla} \ell(\mathbf{w})) = \mathbf{0}.$$

Recall that when $\psi_i(w_i; \varepsilon) \leq 0$ and $-u_i \cdot \psi'_i(w_i; \varepsilon) \leq -\alpha \psi_i(w_i; \varepsilon)$, we have to take the skewed gradient direction $-\alpha \psi_i(w_i; \varepsilon) / \psi'_i(w_i; \varepsilon)$, we denote this direction as $u_i^{(k)}$ hereafter.

Corollary 3 (Convergence of the Pushing Force). Given the assumptions as in Lemma 2, and fix an arbitrary $0 < \varepsilon \leq \inf_{1 \leq i \leq d} \inf_{1 \leq j < K_i} |q_i^{(j)} - q_i^{(j+1)}|^2/4$, where $\{q_i^{(j)}\}$ are the quantization levels. Denote $[q_-, q_+]$ as the set $C_{\varepsilon, i} \cap [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$, where $C_{\varepsilon, i}$ is the projection of C_ε on the i -th coordinate. Let $0 < \varepsilon_1 < (q_i^{(j)} + q_i^{(j+1)} - 2q_+)/3$ and $0 < \varepsilon_2 < (2q_- - q_i^{(j)} - q_i^{(j-1)})/3$ be some small perturbations on a quantization level. Then,

$$(a) \quad \left| \frac{-\alpha \psi_i(q_+ + \varepsilon_1; \varepsilon)}{\psi'_i(q_+ + \varepsilon_1; \varepsilon)} \right| \leq 2\alpha\varepsilon_1;$$

$$(b) \quad \left| \frac{-\alpha \psi_i(q_- - \varepsilon_2; \varepsilon)}{\psi'_i(q_- - \varepsilon_2; \varepsilon)} \right| \leq 2\alpha\varepsilon_2.$$

Moreover, we have $\lim_{k \rightarrow \infty} \|\mathbf{u}^{(k)}\|_2 = 0$ almost surely.

Proof. By symmetry, we only need to prove the statement as in (a). Note that

$$\psi_i(q_+; \varepsilon) = \varepsilon - (q_+ - q_i^{(j-1)})(q_i^{(j)} - q_+) = 0. \quad (9)$$

Also,

$$\psi'_i(w; \varepsilon) = 2w - q_i^{(j+1)} - q_i^{(j)}. \quad (10)$$

Using (10), we expand the numerator of the left-hand side of (a) and obtain

$$\begin{aligned} \psi_i(q_+ + \varepsilon_1; \varepsilon) &= \varepsilon + (q_+ + \varepsilon_1 - q_i^{(j)})(q_+ + \varepsilon_1 - q_i^{(j+1)}) \\ &= \varepsilon + (q_+ - q_i^{(j)})(q_+ - q_i^{(j+1)}) + \varepsilon_1(2q_+ - q_i^{(j+1)} - q_i^{(j)}) + \varepsilon_1^2 \\ &= \varepsilon_1(2q_+ - q_i^{(j+1)} - q_i^{(j)}) + \varepsilon_1^2 < 0. \end{aligned}$$

Since $\varepsilon_1 > 0$, we have $\psi'_i(q_+ + \varepsilon_1; \varepsilon) = 2q_+ + \varepsilon_1 - q_i^{(j+1)} - q_i^{(j)} < 0$ by (11). Then,

$$\begin{aligned} \left| \frac{-\alpha \psi_i(q_+ + \varepsilon_1; \varepsilon)}{\psi'_i(q_+ + \varepsilon_1; \varepsilon)} \right| &= \frac{\alpha \varepsilon_1(2q_+ - q_i^{(j)} - q_i^{(j+1)} + \varepsilon_1)}{(2q_+ - q_i^{(j)} - q_i^{(j+1)}) + 2\varepsilon_1} \\ &\leq \frac{\alpha \varepsilon_1(2q_+ - q_i^{(j)} - q_i^{(j+1)})}{(2q_+ - q_i^{(j)} - q_i^{(j+1)}) + 2\varepsilon_1} \\ &= \alpha \varepsilon_1 - \frac{\alpha \varepsilon_1^2}{(2q_+ - q_i^{(j)} - q_i^{(j+1)}) + 2\varepsilon_1}. \end{aligned}$$

Since $\varepsilon_1 \leq (q_i^{(j)} + q_i^{(j+1)} - 2q_+)/3$, we have

$$\left| \frac{-\alpha \psi_i(q_+ + \varepsilon_1; \varepsilon)}{\psi'_i(q_+ + \varepsilon_1; \varepsilon)} \right| \leq 2\alpha\varepsilon_1.$$

The perturbation ε_1 is then controlled by the convergence of the algorithm to the feasible set, which means $\varepsilon_1 \rightarrow 0$ and it immediately implies that $\mathbf{u} \rightarrow \mathbf{0}$. $\square$

Assumption 5 (Selection of Stepsizes). We choose the sequence of stepsizes $(\gamma_k)_{k \geq 0}$, where $\gamma_k = \gamma_0/k$ and $\gamma_0 \leq 2/L$. This is a special case of Assumption 3.

Theorem 2 (Deterministic Convergence). Assume that Assumptions 2, 5 holds, and $0 < \varepsilon \leq \inf_{1 \leq i \leq d} \inf_{1 \leq j < K_i} |q_i^{(j)} - q_i^{(j+1)}|^2/4$, where $\{q_i^{(j)}\}$ are the quantization levels, then there exists $N \in \mathbb{N}$ such that $\sum_{k=0}^{\infty} (\|\mathbf{v}^{(k+N)}\|_2^2/(k+N)) \rightarrow 0$ almost surely.

Proof. By the definition of limits, we first fix

$$0 < \varepsilon_1 < \min \left\{ \inf_{1 \leq i \leq d} \inf_{1 \leq j < K_i} \frac{q_i^{(j)} + q_i^{(j+1)} - 2q_+}{3}, \inf_{1 \leq i \leq d} \inf_{1 < j \leq K_i} \frac{q_i^{(j)} + q_i^{(j-1)} - 2q_-}{3} \right\},$$

and then we apply Lemma 2 to obtain $N(\varepsilon_1)$ so that $u_i^{(k)} \leq 2\alpha\varepsilon_1$ for all $k \geq N(\varepsilon_1)$ and $i \in [d]$ for the forthcoming argument.

Then, we apply the descent lemma on the L_ℓ -smooth function ℓ . After that, we discuss the two cases of the update coordinate-wisely. We obtain the following inequality:

$$\begin{aligned} \ell(\mathbf{w}^{(k+1)}) &\leq \ell(\mathbf{w}^{(k)}) + \gamma_k (\mathbf{v}^{(k)})^\top \nabla \ell(\mathbf{w}^{(k)}) + \frac{\gamma_k^2 L_\ell}{2} \|\mathbf{v}^{(k)}\|_2^2 \\ &= \ell(\mathbf{w}^{(k)}) + \gamma_k \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} u_i^{(k)} [\nabla \ell(\mathbf{w}^{(k)})]_i - \gamma_k \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 \\ &\quad + \frac{\gamma_k^2 L_\ell}{2} \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2 + \frac{\gamma_k^2 L_\ell}{2} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2. \end{aligned}$$

Since $\gamma_k < 2/L_\ell$, we have $1 - \gamma_k L_\ell/2 > 0$, then

$$\ell(\mathbf{w}^{(k+1)}) \leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma_k}{2} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma_k \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} u_i^{(k)} [\nabla \ell(\mathbf{w}^{(k)})]_i + \frac{\gamma_k^2 L_\ell}{2} \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2.$$

Since the gradient of ℓ is bounded by L_ℓ , and the skewed update direction $\mathbf{u}^{(k)}$ is bounded by $2\alpha\varepsilon_1$, we have for $k \geq N(\varepsilon_1)$ that:

$$\begin{aligned} \ell(\mathbf{w}^{(k+1)}) &\leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma_k}{2} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma_k \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} 2\alpha\varepsilon_1 L_\ell + \frac{\gamma_k^2 L_\ell}{2} \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} (2\alpha\varepsilon_1)^2 \\ &\leq \ell(\mathbf{w}^{(k)}) - \frac{\gamma_k}{2} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \gamma_k n 2\alpha\varepsilon_1 L_\ell + \frac{\gamma_k^2 L_\ell}{2} n 4\alpha^2 \varepsilon_1^2. \end{aligned}$$

Rearranging the inequality and adding the squared skewing term yields

$$\begin{aligned} \frac{\gamma_k}{2} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 &\leq \ell(\mathbf{w}^{(k)}) - \ell(\mathbf{w}^{(k+1)}) + 2\gamma_k n \alpha \varepsilon_1 L_\ell + 2\gamma_k^2 L_\ell n \alpha^2 \varepsilon_1^2 \\ \frac{\gamma_k}{2} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \frac{\gamma_k}{2} \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2 &\leq \ell(\mathbf{w}^{(k)}) - \ell(\mathbf{w}^{(k+1)}) + 2\gamma_k n \alpha \varepsilon_1 L_\ell \\ &\quad + 2\gamma_k^2 L_\ell n \alpha^2 \varepsilon_1^2 + 2\gamma_k n \alpha^2 \varepsilon_1^2 \\ \frac{\gamma_k}{2} \sum_{i \notin T_\varepsilon(\mathbf{w}^{(k)})} [\nabla \ell(\mathbf{w}^{(k)})]_i^2 + \frac{\gamma_k}{2} \sum_{i \in T_\varepsilon(\mathbf{w}^{(k)})} (u_i^{(k)})^2 &\leq \ell(\mathbf{w}^{(k)}) - \ell(\mathbf{w}^{(k+1)}) + D\varepsilon_1, \end{aligned}$$

where D is picked such that $D > 2\gamma_k n \alpha \varepsilon_1 (L_\ell + \gamma_k L_\ell \alpha \varepsilon_1 + \alpha \varepsilon_1)$.

Note that the left hand side of the inequality is simply $\gamma_k/2 \|\mathbf{v}^{(k)}\|_2^2$. Hence,

$$\frac{\gamma_k}{2} \|\mathbf{v}^{(k)}\|_2^2 \leq \ell(\mathbf{w}^{(k)}) - \ell(\mathbf{w}^{(k+1)}) + D\varepsilon_1.$$

Summing over T epochs, we have

$$\sum_{k=0}^{T-1} \frac{\gamma_{k+N}}{2} \|\mathbf{v}^{(k+N)}\|_2^2 \leq \ell(\mathbf{w}^{(N)}) - \ell(\mathbf{w}^{(N+T)}) + TD\varepsilon_1.$$

Further set $\gamma_{k+N} = 1/(L_\ell(k+N))$, then

$$\sum_{k=0}^{T-1} \frac{1}{k+N} \|\mathbf{v}^{(k+N)}\|_2^2 \leq \frac{2L_\ell}{T} \left[\ell(\mathbf{w}^{(N)}) - \ell(\mathbf{w}^{(N+T)}) \right] + 2L_\ell D\varepsilon_1.$$

Hence, when $T \rightarrow \infty$, we have

$$\sum_{k=0}^{T-1} \frac{1}{k+N} \|\mathbf{v}^{(k+N)}\|_2^2 \leq L_\ell D\varepsilon_1.$$

As ε_1 can be arbitrary small (almost surely) depending on N (as N is set such that $\sup d(\mathbf{w}^{(k)}) < \varepsilon_1$ for all $k \geq N$),

$$\sum_{k=0}^{T-1} \frac{1}{k+N} \|\mathbf{v}^{(k+N)}\|_2^2 \rightarrow 0, \text{ as } T \rightarrow \infty.$$

□

We can randomize the output $\mathbf{w}^{(T)}$ and obtain an expected gradient of zero of the loss function, i.e. $\nabla \ell(\hat{\mathbf{w}}^{(T)})$. This is a standard technique in [8] by Gholami and Lan.

6 Experiments

Two Moons Classification. We train a five-layer MLP (shown as below) with a 2-dimensional input, and ReLU as activation function on the non-linear two moon classification task for $T = 60$ epochs. Our dataset consists of $n = 2000$ training samples (in batches of 200 per iteration, points are colored in blue and red) and $m = 500$ test samples (color in black and white), which is generated by `scikit-learn` library's `make_moons` data generator. We have also added a random Gaussian noise with the variance of $\xi = 0.1$ to offset the point from its original position. We apply the logistic loss function in this experiment. The learning rate lr and the hyperparameter for ASkewSGD α are set to 0.2 and 0.2, respectively.

We compare SGD, BinaryConnect, and ASkewSGD to obtain the following results for the two moons classification problem. The first row shows the performance of different optimizers on the original net before quantization, and the second row depicts the quantized model with the quantization set $\mathcal{Q} = \{\pm 1\}^d$.

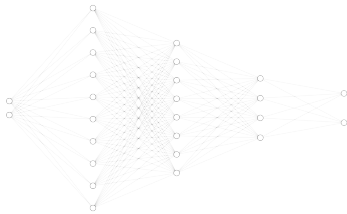


Figure 1: The architecture of the binarized MLP network.

Table 1: Logistic loss after 60 epochs.

Method	Loss	Quantized Loss
Full Precision [W32/A32]	0.00002	0.59982
BinaryConnect [W1/A32]	0.57799	0.03260
ASkewSGD [W1/A32]	0.00367	0.00513

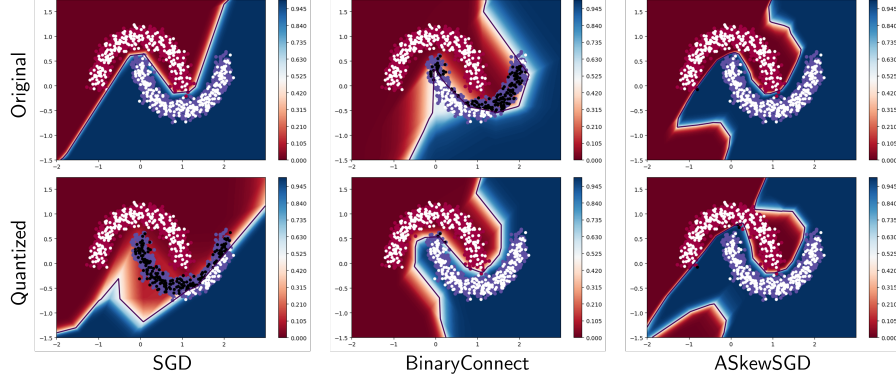


Figure 2: The performance of different optimizers on the original net (top) and the quantized model (bottom).

The original net has its contour completely perturbed by the weight quantization process. BinaryConnect is aware of the quantization scheme, thus generalizing well on the test set, but the original model is completely off from the groundtruth. ASkewSGD, on the other hand, has only a slight skew away from the sensible region, and the quantized model is still able to generalize well on the test set, which demonstrates the strength of this optimization method.

MNIST. We train a binarized MLP with 3 hidden layers, each with 1024 neurons, without bias on the MNIST dataset for $T = 50$ epochs. The dataset consists of $n = 50000$ training samples and $m = 10000$ test samples of handwritten digits images. These images are 28×28 pixels, and we flatten them into 784-dimensional vectors. The training set is divided into batches of size 100. We apply the cross-entropy loss function in this experiment. For SGD, BinaryConnect and ProxQuant, we set the learning rate to 0.001, 0.00001 and 0.001, respectively. For ASkewSGD, we set the learning rate lr and the hyperparameter α to 0.0001 and 0.01, respectively. We divide the learning rate by 2 at epochs 5, 10, 15, 20, 25 and 35 for all methods. Furthermore, ε decays exponentially with $\varepsilon_t = 0.88^{t-25}$, starting from epoch $T = 26$. The hyperparameter λ for ProxQuant is also similarly set to $\lambda_k = 1/(0.88)^{k-15}$, starting from epoch $T = 16$. We apply the cross-entropy loss function in this experiment.

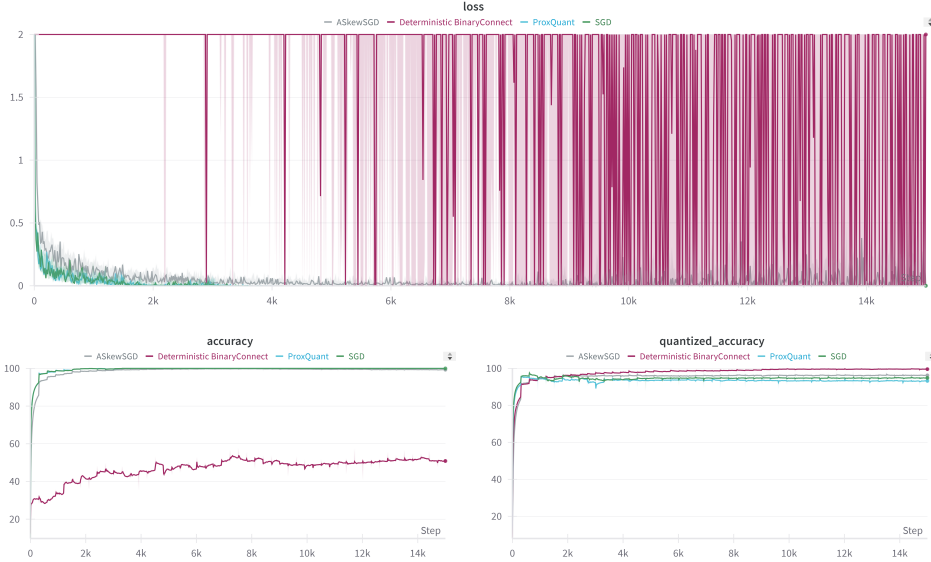


Figure 3: The performance of different optimizers, measured by the non-quantized loss, accuracy, and accuracy after quantization (quantized accuracy).

Table 2: Performance of A Simple Multi-Layer Perceptron on MNIST after 50 epochs.

Method	Accuracy	Quantized Accuracy	Loss
Full Precision [W32/A32]	98.67%	93.19%	6×10^{-8}
ASkewSGD [W1/A32]	97.91%	94.95%	3×10^{-8}
BinaryConnect [W1/A32]	49.55%	95.79%	3.80773
ProxQuant [W1/A32]	98.53%	91.39%	1×10^{-6}

We are not able to observe a significant difference in the performance of ASkewSGD and BinaryConnect on the MNIST dataset. However, the BinaryConnect method fluctuates severely when the loss is optimized. This is often undesired in practice, as the model may not be able to generalize well on the test set. Nevertheless, ASkewSGD outperforms ProxQuant in terms of quantized accuracy. We even observe surprisingly high accuracy of when using the plain SGD method, demonstrating the simplicity of the learning task.

Computer Vision Task. We train ResNet-18 without bias and tunable parameters on the batch normalization layer on the CIFAR-10 dataset for $T = 150$ epochs. The dataset consists of $n = 50000$ training samples and $m = 10000$ test samples. We apply the cross-entropy loss function in this experiment. We set the learning rate lr and the hyperparameter for ASkewSGD α to 0.06 and 0.2, respectively. The quantization set $\mathcal{Q}$ is set to $\{\pm 1\}^d$.



Table 3: Performance of ResNet-18 on CIFAR10 after 60 epochs.

Method	Accuracy	Loss	Quantized Loss
Full Precision [W32/A32]	88.91%	0.04533	0.1005
ASkewSGD [W1/A32]	86.91%	0.12337	0.8669
BinaryConnect [W1/A32]	79.46%	0.56677	1.9637

The quantized ASkewSGD has its accuracy dropped for only two points when compared to the full precision model, which shows its superiority. It is worthwhile to note that the accuracy of the quantized model skyrocketed to 50% only after running for 40 epochs, meaning that quantized models are sensitive to the accumulation of positional errors and the uncertain landscape of the loss function. Compared to the MNIST dataset, the difference between the ASkewSGD and BinaryConnect methods is more pronounced. Due to the limited time, we leave the 4-bit quantization and the comparison for future work.

7 Summary

In this report, we have presented a new optimization method called ASkewSGD for training quantized neural networks. We have shown that the algorithm converges to a stationary point of the loss function with full-batch gradient with assumptions on the Lipschitzness of the loss function and decreasing stepsizes. We have also provided experimental results on several datasets, including two moons classification, MNIST, and CIFAR-10, demonstrating the effectiveness of our presented method compared to existing methods. Future works will continue to address the convergence of ASkewSGD under the presence of noise in the stochastic case.

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Appendix

A Proof of Lemma 1

We reproduce the fundamental connection between the Clarke's tangent cone and the quantization constraints, for educational purposes. The fact that we do not have equality constraints in our case simplifies the proof.

Lemma 1. Let $\mathbf{x} \in \mathbb{R}^d$ be a point in the feasible set C . Then, the Clarke's tangent cone $T(\mathbf{x})$ is equal to the quantization cone $T_M(\mathbf{x})$ defined in (6).

Proof. Adapted from R. Herzog, [5].

$(\Rightarrow)$: $\delta\mathbf{x} \in T(\mathbf{x})$ implies that there exists two sequences $\{\mathbf{x}_j\} \rightarrow \mathbf{x}, \{t_j\} \subset C, t_j \downarrow 0$ for all $j \in \mathbb{N}$ and

$$\frac{\mathbf{x}_j - \mathbf{x}}{t_j} \rightarrow \delta\mathbf{x}.$$

Let us fix $i \in I(\mathbf{x})$, then by the Mean-Value Theorem,

$$\begin{aligned} 0 &\leq g_i(\mathbf{x}_k) \text{ since } \mathbf{x}_k \text{ is feasible} \\ &= g_i(\mathbf{x}) + \nabla g_i(\mathbf{x} + \xi_j(\mathbf{x}_j - \mathbf{x}))^\top (\mathbf{x}_j - \mathbf{x}) \\ &= 0 + \nabla g_i(\mathbf{x} + \xi_j(\mathbf{x}_j - \mathbf{x}))^\top (\mathbf{x}_j - \mathbf{x}) \end{aligned}$$

for some $\xi_j \in (0, 1)$.

Divide both sides by t_j and take the limit as $j \rightarrow \infty$, we have $0 \leq \nabla g_i(\mathbf{x})^\top \delta\mathbf{x}$ wherever $i \in I(\mathbf{x})$.

$(\Leftarrow)$: Let $\delta\mathbf{x} \in T_M(\mathbf{x})$ satisfy $\nabla g_i(\mathbf{x})^\top \delta\mathbf{x} \geq 0, \forall i \in I(\mathbf{x})$, also let $\delta\mathbf{y}$ be given by MFCQ such that $\nabla g_i(\mathbf{w})^\top \delta\mathbf{y} > 0, \forall i \in I(\mathbf{x})$. Put $\tau(t) := \delta\mathbf{x} + t \cdot \delta\mathbf{y}$. Then for all $t > 0$, we have $\nabla g_i(\mathbf{x})^\top \tau(t) > 0, \forall i \in I(\mathbf{x})$, implying that $\tau(t)$ are all feasible MFCQ vectors.

Now, we claim that $\tau(t) \in T(\mathbf{x})$ for all $t \in \mathbb{R}_{++}$. Let $\gamma(u) := \mathbf{x} + u\tau(t)$, $u \in (-\varepsilon, \varepsilon)$, for an infinitesimally small ε , given by the continuity of g . Then, $\gamma(u) \in T_M$ for every $u \in [0, \varepsilon)$ and $\gamma(0) = \mathbf{x}, \gamma'(0) = \tau(t)$. For an arbitrary sequence $\{u_j\} \downarrow 0$ and $\mathbf{x}_k = \gamma(u_j) \rightarrow \mathbf{x}$ we have

$$\tau(t) = \gamma'(0) = \lim_{j \rightarrow \infty} \frac{\gamma(u_j) - \gamma(0)}{u_j - 0} = \lim_{j \rightarrow \infty} \frac{\mathbf{x}_j - \mathbf{x}}{t_j} \in T(\mathbf{x}).$$

Since $T(\mathbf{x})$ is closed, $\delta\mathbf{x} = \lim_{t \rightarrow 0} \tau(t) \in T(\mathbf{x})$. □

B Proof of Lemma 2

Lemma 2 (Leconte et al., [1]). Under Assumptions 2-3, it holds that $\limsup_{k \rightarrow \infty} d(\mathbf{w}_k, C_\varepsilon) = 0$ almost surely.

Proof. Denote $M_1 = \max\{M_c, L_\ell\}$. Now, for all $i \in [n]$, we have $\|\widehat{\nabla} \ell(\mathbf{w}^{(k)})\|_2 \leq M_1$ and $|v_i^{(k)}| \leq M_1$, and thus $|w_i^{(k+1)} - w_i^{(k)}| \leq \gamma_k M_1$.

Three crucial steps for the correctness of this lemma will then be introduced. The three steps involved will correspond to Claims 1-3, Claims 4-5, and Claims 7-8.

Claim 1. For $i \in \{1, 2, \dots, d\}$, and for $2 \leq j \leq K_i - 1$ if the set $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$ is visited by $w_i^{(k)}$ infinitely often, then there is k_0 such that for all $k > k_0$, $w_i^{(k)} \in [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$.

Fix a j and denote $[q_-, q_+]$ the set $C_{\varepsilon, i} \cap [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$, where $C_{\varepsilon, i}$ is the projection of C_ε on the i -th coordinate. Define

$$k_0 = \sup\{k : \gamma_k M_1 \geq \max(q_- - (q_j^i + q_{j-1}^i)/2, (q_j^i + q_{j+1}^i)/2 - q_+)\}.$$

Consider $k \geq k_0$, suppose $(q_i^{(j)} + q_i^{(j-1)})/2 \leq w_i^{(k)} < q_-$ (the left side of the interval), then the iterate is pushed to the right and $w_i^{(k)} \leq w_i^{(k+1)}$.

Furthermore, by definition of k_0 , it holds that $w_i^{(k+1)} \leq q_- + \gamma_k M_1 \leq (q_i^{(j)} + q_i^{(j+1)})/2$. This implies that, in this case $w_i^{(k+1)}$ stays in $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$.

Otherwise, suppose $q_+ \leq w_i^{(k+1)} < (q_i^{(j)} + q_i^{(j+1)})/2$ (the right side of the interval), then the iterate is pushed to the left and $w_i^{(k)} \geq w_i^{(k+1)} \geq q_+ - \gamma_k M_1 \geq (q_i^{(j)} + q_i^{(j-1)})/2$. Finally, if $w_i^{(k)} \in [q_-, q_+]$, then by the definition of k_0 , we obtain $w_i^{(k+1)} \in [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$.

Thus, we have shown that for $k \geq k_0$ if $w_i^{(k)}$ stays within the interval $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$, then for all $k' \geq k$, $w_i^{(k')} \in [(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$.

Similarly, we can prove the two statements below corresponding to the edge cases.

Claim 2. For $i \in \{1, 2, \dots, d\}$, if the set $(-\infty, (q_i^{(1)} + q_i^{(2)})/2)$ is visited by $w_i^{(k)}$ infinitely often, then there is k_0 such that for all $k > k_0$, $w_i^{(k)} \in (-\infty, (q_i^{(1)} + q_i^{(2)})/2)$.

Claim 3. For $i \in \{1, 2, \dots, d\}$, if the set $[(q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$ is visited by $w_i^{(k)}$ infinitely often, then there is k_0 such that for all $k > k_0$, $w_i^{(k)} \in [(q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$.

We can then leave our attention to one of the three intervals $[(q_i^{(j)} + q_i^{(j-1)})/2, (q_i^{(j)} + q_i^{(j+1)})/2]$, $(-\infty, (q_i^{(1)} + q_i^{(2)})/2)$, or $[(q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$.

Claim 4. There is $k_1 \geq k_0$, such that if there are two indices $m_+ \geq m_- > k_0$ such that $w_i^{(m_-)} < q_- < q_+ < w_i^{(m_+)}$, then there is m , satisfying $m_- \leq m \leq m_+$, such that $w_i^{(m)} \in [q_-, q_+]$.

Define $k_1 = \sup\{k : \gamma_k M_1 \geq q_+ - q_-\}$. Let m_-, m_+ be given and consider $m = \inf\{k \geq m_- : w_i^{(k)} \geq q_-\}$. It holds that $w_i^{(m-1)} < q_- \leq w_i^{(m)} \leq w_i^{(m-1)} + \gamma_k M_1$.

Since $m \geq k_1$, this implies that $w_i^{(m)} \leq q_- + \gamma_k M_1 \leq q_+$, which proves the claim.

Claim 5. There is $k_1 \geq k_0$, such that if there are two indices $m_- \geq m_+ > k_0$ such that $w_{m_-} < q_- < q_+ < w_{m_+}$, then there is m , satisfying $m_+ \leq m \leq m_-$, such that $w_m^i \in [q_-, q_+]$.

The proof of Claim 5 is similar to Claim 4. Claims 4, and 5 show that there are only three behaviors of $w_i^{(k)}$. These three conditions will be treated by Claims 6, 7, and 8, respectively.

Claim 6. Let $k \geq k_0$. If $w_i^{(k)}$ visits $[q_-, q_+]$ infinitely often, then $\limsup_{k \rightarrow \infty} w_i^{(k)} \leq q_+$ and $\limsup_{k \rightarrow \infty} w_i^{(k)} \geq q_-$.

We first show the case of $\limsup_{k \rightarrow \infty} w_i^{(k)} \leq q_+$. Suppose that $w_i^{(k)} > q_+$, then $w_i^{(k+1)} \leq w_i^{(k)}$, and otherwise if $w_i^{(k)} \leq q_+$ then $w_i^{(k+1)} \leq q_+ + \gamma_k M_1$. We note that γ_k is non-increasing and thus no iterates $w^{(k')}$ where $k' \geq k$ can go further than $q_+ + \gamma_k M$ since it immediately receives the pushing force after the constraint violation. Tending k to infinity yields $\limsup_{k \rightarrow \infty} w_k^i \leq c_+$. It is easy to see that the other case also holds.

Claim 7. If for all $k \geq k_0$ large enough, $w_i^{(k)} > q_+$, then $w_i^{(k)} \rightarrow q_+$.

For all k large enough, the sequence w_k^i receives a pushing force from the right and thus is decreasing. From the given condition that $w_i^{(k)} > q_+$, $w_i^{(k)}$ is decreasing and bounded below, thus the sequence has a limit. Assume the contrary that $\lim_{k \rightarrow \infty} w_i^{(k)} \neq q_+$. It holds that $w_i^{(k+m+1)} \leq w_i^{(k)} - M_+ \sum_{j=0}^m \gamma_{k+j}$, where $M_+ = \inf\{\min(M_{\varepsilon,c}, \alpha|\psi_i(w_i; \varepsilon)|/\psi_i'(w_i; \varepsilon)) : w_i \in [q_+, (q_i^{(j)} + q_i^{(j+1)})/2]\} > 0$. Since $\sum_{j=0}^{\infty} \gamma_j = +\infty$, we have reached a contradiction. $w_i^{(k)} \rightarrow q_+$.

Claim 8. If for all $k \geq k_0$ large enough, $w_i^{(k)} < q_-$, then $w_i^{(k)} \rightarrow q_-$.

It holds that $w_i^{(k+m+1)} \geq w_i^{(k)} + M_- \sum_{j=0}^m \gamma_{k+j}$, where $M_- = \inf\{\min(M_{\varepsilon,c}, \alpha|\psi_i(w_i; \varepsilon)|/\psi_i'(w_i; \varepsilon)) : w \in ((q_i^{(j)} + q_i^{(j-1)})/2, q_-]\} > 0$. Similar to Claim 7, it is impossible that $\lim_{k \rightarrow \infty} w_i^{(k)} \neq q_-$. $\square$